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Paper Title

Paper Title: Hierarchical Transformer Preconditioning for Interactive Physics Simulation

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Abstract

Abstract (Maximum 300 words):

Neural preconditioners for real-time physics simulation offer promising data-driven priors, but they often fail to capture long-range couplings efficiently because they inherit local message passing or sparse-operator access patterns. We introduce the Hierarchical Transformer Preconditioner, a neural preconditioner anchored to a weak-admissibility H-matrix partition. The partition

provides a multiscale structural prior (dense diagonal leaves plus coarsening off-diagonal tiles) that enables full-graph approximate-inverse computation with $O(N)$ scaling at fixed block sizes. The network models the inverse through low-rank far-field factors and uses highway connections (axial buffers plus a global summary token) to propagate context across transformer depth. At each PCG iteration, preconditioner application reduces to batched dense GEMMs with regular memory access.

The key training contribution is a cosine-Hutchinson probe objective that learns the action of MA on convergence-critical spectral subspaces, optimizing angular alignment of MAz with z rather than forcing eigenvalue clusters to a prescribed location. This removes unnecessary spectral-placement constraints from SAI-style objectives and improves conditioning on irregular spectra. Because both inference and apply are dense, dependency-free tensor programs, the full solve loop is captured as a single CUDA Graph.

On stiff multiphase Poisson systems (up to 100 : 1 density contrast, $N = 1,024$ – $16,384$), the solver runs from ~ 143 to ~ 21 fps. At $N = 8,192$, it reaches 17.9 ms/frame, with $2.2\times$ speedup over GPU Jacobi, $\sim 28\times$ over GPU IC/DILU (AMGX multicolor_dilu), and $2.7\times$ over neural SPAI retrained per scale on the same benchmark.

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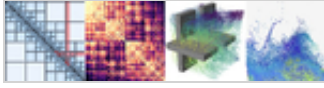
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